Sales Demand Forecasting for

Inventory Planning

FreshMart — FMCG Retail Chain | Supply Chain Analytics | Time Series | Regression

April 2025

1. Executive Summary

FreshMart manages thousands of SKUs across multiple store locations and currently relies on simple historical averages for inventory planning — resulting in frequent stockouts and costly overstocking. This project builds a demand forecasting system using Linear Regression and Random Forest Regressor on 731 days of sales data. The best model achieved MAPE of 9.1% and R2 of 0.816, and generated a 30-day forward forecast recommending 3,861 units of total stock.

2. Problem Definition

Business Problem

• Frequent stockouts of high-demand products causing revenue loss

• Overstocking of slow-moving items increasing holding and wastage costs

• Demand planning based on simple historical averages — not capturing seasonality or trends

• No product-level or store-level forecasting capability

• Promotions and holidays not factored into existing inventory decisions

Key Objectives

• Analyse sales trends, seasonality, and weekly/monthly demand patterns

• Quantify the impact of promotions, holidays, and pricing on demand

• Build an accurate forecasting model with MAPE below 15%

• Compare at least two models and select best based on error metrics

• Generate a 30-day forward demand forecast

• Provide actionable inventory planning recommendations

3. Dataset Description

Dataset Attribute Details

Total Records 731 daily sales records (2 years)

Total Features 15 input features + 1 target

Products (SKUs) 10 products (SKU_100 to SKU_109)

Stores 5 stores (STORE_1 to STORE_5)

Categories Beverages, Groceries, Dairy, Snacks

Key Features Price, Discount, Promotion_Flag,

Holiday_Flag, Season, Stock_Available

Target Variable Units_Sold (daily)

Missing Values None — dataset fully clean

4. Methodology & Data Preprocessing

Data Preprocessing

• Date column converted to datetime format for time-based feature extraction

• Label Encoding applied to: Store_ID, Product_ID, Category, Season

• Feature Engineering: extracted Week_of_Year and Quarter from Date

• Train/Test split: first 80% for training, last 20% for testing (time-order preserved — no data leakage)

• StandardScaler applied to all features

Modelling Approach

• Model 1 — Linear Regression: simple interpretable baseline

• Model 2 — Random Forest Regressor (100 trees): captures non-linear patterns

• Evaluation metrics: MAE, RMSE, MAPE, R2 Score

• Feature importance extracted from Random Forest

• 30-day future forecast generated using average feature values as input

• 10% confidence band added to forecast for safety stock planning

5. Results & Model Performance

KPI / Metric Target / Benchmark

Linear Regression — MAE 7.52 units

Linear Regression — MAPE 11.8%

Linear Regression — R2 0.742

Random Forest — MAE 5.89 units

Random Forest — MAPE 9.1% (below 15% target)

Random Forest — R2 0.816 (explains 81.6% of variance)

Promotion Sales Lift +33.9% more units during promotion periods

30-Day Forecast Total 3,861 units (~129 units/day average)

Recommended Stock Order 4,247 units (+10% safety buffer)

Demand Insights

Finding Detail

Top selling product SKU_107 — needs priority restocking

Top category Dairy — highest average daily demand

Highest demand season Summer — stock up before season begins

Top demand driver Promotion_Flag (33.9% importance score)

6. Business Recommendations

• Replace manual average-based planning with this ML forecast model immediately

• Order stock at least 7 days before high-demand months to avoid stockouts

• Safety stock formula: Average Daily Demand x 1.5 for top-selling SKUs

• Total stock to order for next 30 days: 4,247 units (+10% buffer over forecast)

• Schedule promotions strategically — they increase demand by 33.9%

• Reduce order quantities for bottom-3 products by 20% to cut overstock costs

• Stock up before Summer season — highest average demand across all products

• Retrain model monthly with fresh sales data to maintain forecast accuracy

• Build store-manager dashboard showing daily forecast vs actual units

7. Conclusion & Future Work

• Random Forest (MAPE 9.1%, R2 0.816) significantly outperforms Linear Regression baseline

• Promotion_Flag is the single most important demand driver at 33.9% feature importance

• The model is ready to replace manual historical-average planning at FreshMart

• Future work: ARIMA/SARIMA for pure time series, product-level individual models, LSTM for deep learning

• Target: reduce stockouts by 20% and overstock holding costs by 15% within 6 months